

Program Evaluation and Quasi Experiments

Chris Conlon

Wednesday 5th November, 2025

Applied Econometrics

Recall: Fundamental Problem of Causal Inference

Fundamental Problem of Causal Inference

- ▶ We don't observe the **counterfactual** $Y_i(1 - D_i)$.
- ▶ For a single individual we either observe $Y_i(1)$ **or** $Y_i(0)$ but never both!
 - ex: We don't see what your wage would have been if you didn't attend college.
 - ex: We might know your cholesterol before you took Lipitor, but we don't know what it would be today if you didn't take Lipitor.

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

i	$Y_i(1)$	$Y_i(0)$	D_i	X_i
1	1	?	1	3.2
2	0	?	1	1.7
3	?	0	0	2.5
		$\vdots$		
n	?	1	0	-1.5

How do we construct the counterfactual?

We observe $Y_i(D_i)$ but not $Y_i(1 - D_i)$, how do we estimate it?

Event Study: Compare $Y_{t+k}(1)$ to $Y_{t-j}(0)$ before and after an event at t .

Matching: Compare $Y_i(1)$ to $Y_j(0)$ for $i \neq j$ but where $X_i = X_j$.

Difference in Difference: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $Y_{j,t}(0) - Y_{j,t-1}(0)$ for some $j \neq i$.

Synthetic Control: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $\bar{Y}_{-i,t}(0) - \bar{Y}_{-i,t-1}(0)$ for some average over $j \neq i$ so that $X_{i,t} \approx \bar{X}_{-i,t}$.

Matching

Matching

- ▶ Compare treated individuals to un-treated individuals with identical observable characteristics X_i .
- ▶ Key assumption: everything about $Y_i(1) - Y_i(0)$ is captured in X_i ; or u_i is randomly assigned conditional on X_i .
- ▶ Basic idea: The treatment group and the control group don't have the same distribution of observed characteristics as one another.
- ▶ **Re-weight** the un-treated population so that it resembles the treated population.
- ▶ Once distribution of X_i is the same for both groups $X_i|D_i \sim X_i$ then we assume all other differences are irrelevant and can just compare means.
- ▶ Matching assumes **all selection is on observables**.

- Formally the key assumption is the **Conditional Independence Assumption (CIA)**

$$\{Y_i(1), Y_i(0)\} \perp D_i | X_i$$

- Once we know X_i allocation to treatment D_i is as if it is random.
- The only difference between treatment and control is composition $f(X_i | D_i)$ of the sample.

Nonparametric k -NN Matching: Abadie and Imbens (2002)

For each observation D_i , we observe $Y_i(D_i)$ compute a **counterfactual** $\hat{Y}_i(1 - D_i)$:

$$\begin{aligned}\hat{Y}_i(0) &= \begin{cases} Y_i & \text{if } D_i = 0 \\ \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} Y_l & \text{if } D_i = 1 \end{cases} \\ \hat{Y}_i(1) &= \begin{cases} \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} Y_l & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases}\end{aligned}$$

- ▶ $\#\mathcal{J}_M(i)$ is the number of matches for i of opposite treatment assignment $D_l = 1 - D_i$.
- ▶ M is the “number of matches” within some distance of $|X_l - X_i| < d_M(i)$.
- ▶ If there are ties $\#\mathcal{J}_M(i) > M$.
- ▶ This is just k -NN matching.

Nonparametric k -NN Matching: Abadie and Imbens (2002)

Each observation i gets a weight based on how often it is used as a match for other observations l :

$$K_M(i) = \sum_{l=1}^N 1\{i \in \mathcal{J}_M(l)\} \frac{1}{\#\mathcal{J}_M(l)}$$

Observations used in lots of matches get more weight. $\sum_{i=1}^N K_M(i) = N$.

This is just a weighted average of Y_i values (aka a **kernel**!):

$$ATE_M = \frac{1}{N} \sum_{i=1}^N [\widehat{Y}_i(1) - \widehat{Y}_i(0)] = \frac{1}{N} \sum_{i=1}^N \underbrace{(2D_i - 1) [1 + K_M(i)]}_{w_{i,M}} Y_i$$

Different weighting schemes give different parameters:

$$ATE_M = \frac{1}{N} \sum_{i=1}^N (2D_i - 1) [1 + K_M(i)] \cdot Y_i$$

$$ATT_M = \sum_{i=1}^{N_1} [D_i - (1 - D_i) \cdot K_M(i)] \cdot Y_i$$

$$ATUT_M = \sum_{i=1}^{N_0} [D_i \cdot K_M(i) - (1 - D_i)] \cdot Y_i$$

Why does any of this work?

Let $F^1(x)$ be the distribution of characteristics in the treatment group, we can define the ATT as

$$\begin{aligned}\mathbb{E}[Y(1) - Y(0) \mid D = 1] &= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(1) - Y(0) \mid D = 1, X)] \\ &= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(1) \mid D = 1, X)] - \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid D = 1, X)] \text{ linearity}\end{aligned}$$

The first part we observe directly:

$$= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(1) \mid D = 1, X)]$$

But the counterfactual mean is not observed!

$$= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid D = 1, X)]$$

But conditional independence does this for us:

$$\mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid D = 1, X)] = \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid D = 0, X)]$$

Nonparametric k -NN Matching: Bias Correction

We can use weighted least squares to adjust the predictions $\hat{Y}_i(D_i)$:

$$(\hat{\beta}_{t,0}, \hat{\beta}_{t,1}) = \operatorname{argmin}_{\{\beta_{t,0}, \beta_{t,1}\}} \sum_{i: T_i=t} K_M(i) (Y_i - \beta_{t,0} - \beta'_{t,1} X_i)^2$$

Where $\hat{\mu}_1(X_i), \hat{\mu}_0(X_i)$ are the regression functions for treatment and control.

$$\begin{aligned} \tilde{Y}_i(0) &= \begin{cases} Y_i & \text{if } D_i = 0 \\ \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} \{Y_l + \hat{\mu}_0(X_i) - \hat{\mu}_0(X_l)\} & \text{if } D_i = 1 \end{cases} \\ \tilde{Y}_i(1) &= \begin{cases} \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} \{Y_l + \hat{\mu}_1(X_i) - \hat{\mu}_1(X_l)\} & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases} \end{aligned}$$

So that the ATE is given by: $ATE_M = \frac{1}{N} \sum_{i=1}^N \{\tilde{Y}_i(1) - \tilde{Y}_i(0)\}$

What about higher dimensions?

- ▶ We know that nearest neighbor is **cursed** in high dimensions.
 - Usual caveats apply: may be doing **extrapolation**.
 - Even more reason to use regression/bias adjustment.
- ▶ Given two vectors $\mathbf{x}$ and $\mathbf{y}$, how to choose $d(\mathbf{x}, \mathbf{y})$ the **distance** function?
- ▶ Papers mostly use **Mahalanobis distance**: $d(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})S^{-1}(\mathbf{x} - \mathbf{y})'$.
 - Quadratic distance with inverse covariance matrix as “weights”.
 - Generalizes Euclidean distance (diagonal S).
- ▶ Older papers use **caliper matching** anything within $\|\mathbf{x}_s - \mathbf{x}_t\| < b$ is match
 - Now number of matches varies from observation to observation.
 - Variance can be unpredictable: some $(\mathbf{y}_i, \mathbf{x}_i)$ have many of matches, others have none
 - Some obs may have nothing within $\|\mathbf{x}_s - \mathbf{x}_t\| < b_w$? Drop these?
 - Probably avoid this unless you have a good reason...

- ▶ Base case is easy enough to do on your own using a canned k -NN package.
- ▶ Even regression/bias adjustment is pretty easy
- ▶ Computing ATE, ATT, ATUT is relatively straightforward on the matched sample
- ▶ MatchIt in R (by Gary King et. al) or nnmatch in Stata (by Abadie and Imbens et al)
- ▶ Documentation here is pretty good:

https://scholar.harvard.edu/files/imbens/files/sjpdf-1.html_.pdf.

Matching in Practice

How do we actually do this?

- ▶ For each entry in the treatment (y_i, x_i)
 - Find all x_s from the control group that is “close enough” $\|x_s - x_t\| < b_w$.
 - For each treated observation compute $\beta(x_t) = y_t - \mathbb{E}[y_s \mid I(\|x_s - x_t\| < b_w)]$.
 - Compute the mean of $\beta(x_t)$
- ▶ Some pitfalls
 - Variance can be unpredictable: some (y_t, x_t) have many of matches, others have none
 - For some (y_t, x_t) may have nothing within $\|x_s - x_t\| < b_w$? Drop these?

Matching in Practice: Inverse Probability Weighting

How do we actually do this?

- Calculate a smoothed estimate of the treatment probability $\pi(x) = \Pr(D_i = 1 \mid x)$.

$$\frac{1}{n} \sum_{t \in \text{Treatment}} \frac{y_t}{\pi(x_t)} - \frac{1}{n} \sum_{s \in \text{Control}} \frac{y_s}{1 - \pi(x_s)}$$

- How to get $\pi(x)$? Run a logit or probit.
- We can stabilize the weights replace $w(x) = \frac{1}{\pi(x)}$ with:

$$w(x) = \frac{\Pr(D = 1)}{\pi(x)} \text{ for } D_i = 1 \quad w(x) = \frac{\Pr(D = 0)}{1 - \pi(x)} \text{ for } D_i = 0$$

- This sometimes helps crazy big weights when treated group is small.

So matching works great in dimension 1. But what if $\dim(X) > 1$?

- ▶ True high-dimensional matching may be infeasible. There may be no set of weights such that:
$$f(X_i \mid D_i = 1) \equiv \int w_i f(X_i \mid D_i = 0) \partial w_i.$$
- ▶ One solution is the nearest-neighbor approach in Abadie Imbens (2006).
- ▶ This is still cursed in that our nearest neighbors get further away as the dimension grows.
- ▶ Suppose instead we had a **sufficient statistic**

- ▶ Rosenbaum and Rubin propose the **propensity score**

$$\Pr(D_i = 1 \mid X_i) \equiv P(X_i)$$

- ▶ They prove that the propensity score and any function of X , $b(X)$ which is finer serves as a **balancing score**.
- ▶ Finer implies that:

$$\begin{aligned} b(X^1) = b(X^2) &\implies P(X^1) = P(X^2) \\ P(X^1) = P(X^2) &\nRightarrow b(X^1) = b(X^2) \end{aligned}$$

- Main result: If treatment assignment is strongly ignorable conditional on X (CIA) then it is strongly ignorable $Y(1), Y(0) \perp D|X$ given any balancing score $b(X)$ including the propensity score:

$$\begin{aligned}\Pr(D = 1|Y(1), Y(0), P(X)) &= \mathbb{E}[\Pr(D = 1|Y(1), Y(0), X)|P(X)] \\ &= \mathbb{E}[Pr(D = 1|x)|P(X)] = P(X)\end{aligned}$$

- Also we require that $0 < P(X) < 1$ at each X which is known as the **support condition**.
- The theorem implies that given $P(X)$ we have as if random assignment.

- ▶ Instead of matching on K dimensional X we can now match on a one-dimensional propensity score
- ▶ Thus the propensity score provides **dimension reduction**
- ▶ We still have to estimate the propensity score which is a high dimensional problem without *ad-hoc* parametric restrictions.
- ▶ Let us begin by assuming a can-opener.
- ▶ An easy way would be to use $\pi(x)$ from logit or probit.

Just like in the matching case the problem arises because we do not observe the counterfactual mean:

$$\mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid T = 1, X)]$$

With conditional independence and the propensity score:

$$\begin{aligned}\mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid T = 1, X)] &= \mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid T = 0, X)] \\ &= \mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid T = 0, P(X))]\end{aligned}$$

- ▶ Think about

$$Y = \alpha + \beta D_i + u$$

- ▶ Assume that $\mathbb{E}(u|D, X) = \mathbb{E}(u|X)$ which is a conditional mean independence assumption
- ▶ Then we can get β consistently (but not other variables) by running the following:

$$Y = \alpha + \beta D_i + \gamma X + v$$

- ▶ Again we are in the homogenous treatment world

How do we implement?

- Kernels are an obvious choice

$$\widehat{ATT} = \frac{1}{N_1} \sum_{i \in T=1} \left[Y_i - \frac{\sum_{j \in T=0} Y_j \cdot \mathcal{K}(P(X_i) - P(X_j))}{\sum_{s \in T=0} \mathcal{K}(P(X_i) - P(X_s))} \right]$$

where N_1 is the sample size of the treatment group

and $K(u)$ is a valid Kernel weight (people tend to use Gaussian Kernels here)

- As your propensity score gets further away from observation i you get less weight
- As $h \rightarrow 0$ (or σ_h) the window gets smaller and we use fewer neighbors.

Case Study

A Matching Example

Here is an example where I found that matching was helpful in my own work with Julie Mortimer:

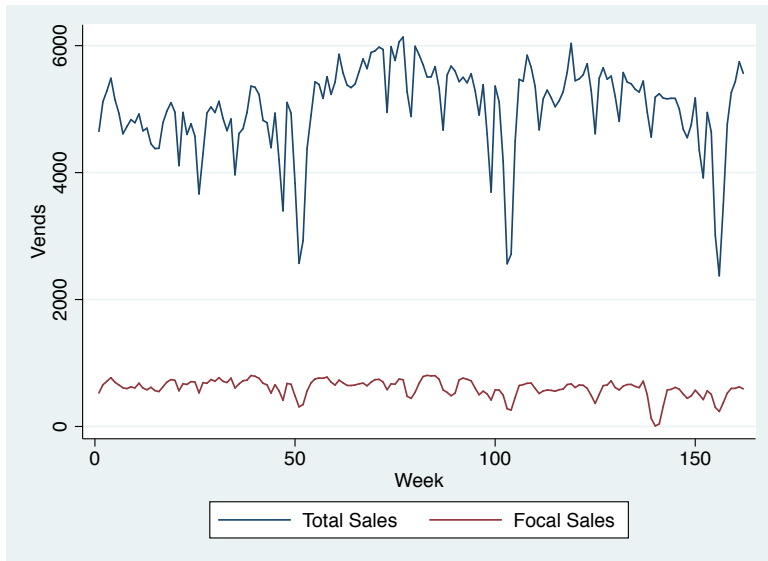
- ▶ We ran a randomized experiment where we removed Snickers bars from around 60 vending machines in office buildings in downtown Chicago.
- ▶ We have a few possible control groups:
 1. Same vending machine in other weeks (captures heterogeneous tastes in the cross section)
 2. Other vending machines in the same week (might capture aggregate shocks, ad campaigns, etc.)
- ▶ We went with #1 as #2 was not particularly helpful.

A Matching Example

Major problem was that there was a ton of heterogeneity in the overall level of (potential) weekly sales which we call M_t .

- ▶ Main source of heterogeneity is how many people are in the office that week, or how late they work.
- ▶ Based on total sales our average over treatment weeks was in the 74th percentile of all weeks.
- ▶ This was after removing a product, so we know sales should have gone down!
- ▶ How do we fix this without running the experiment for an entire year!
- ▶ Can't use shares instead of quantities. Why?

Sales over Time



A Matching Example

Ideally we could just observe M_t directly and use that as our matching variable X

- ▶ We didn't observe it directly and tried a few different measures:
 - Sales at the soda machine next to the snack machine
 - Sales of salty snacks at the same machine (not substitutes for candy bars).
 - We used k-NN with $k = 4$ to select control weeks – notice we re-weight so that overall sales are approximately same (minus the removed product).
- ▶ We also tried a more structured approach:
 - Define controls weeks as valid IFF
 - Overall sales were weakly lower
 - Overall sales were not less than Overall Sales less expected sales less Snickers Sales.

Product	Control Mean	Control %ile	Treatment Mean	Treatment %ile	Mean Difference	% Δ
<i>Vends</i>						
Peanut M&Ms	359.9	73.6	478.3*	99.4	118.4*	32.9
Twix Caramel	187.6	55.3	297.1*	100.0	109.5*	58.4
Assorted Chocolate	334.8	66.7	398.0*	95.0	63.2*	18.9
Assorted Energy	571.9	63.5	616.2	76.7	44.3	7.8
Zoo Animal Cracker	209.1	78.6	243.7*	98.1	34.6*	16.5
Salted Peanuts	187.9	70.4	216.3*	93.7	28.4	15.1
Choc Chip Famous Amos	171.6	71.7	193.1*	95.0	21.5*	12.5
Ruger Vanilla Wafer	107.3	59.7	127.9	78.6	20.6*	19.1
Assorted Candy	215.8	43.4	229.6	60.4	13.7	6.4
Assorted Potato Chips	279.6	64.2	292.4*	66.7	12.8	4.6
Assorted Pretzels	548.3	87.4	557.7*	88.7	9.4	1.7
Raisinets	133.3	66.0	139.4	74.2	6.1	4.6
Cheetos	262.2	60.1	260.5	58.2	-1.8	-0.7
Grandmas Choc Chip	77.9	51.3	72.5	37.8	-5.4	-7.0
Doritos	215.4	54.1	203.1	39.6	-12.3*	-5.7
Assorted Cookie	180.3	61.0	162.4	48.4	-17.9	-10.0
Skittles	100.1	62.9	75.1*	30.2	-25.1*	-25.0
Assorted Salty Snack	1382.8	56.0	1276.2*	23.3	-106.7*	-7.7
Snickers	323.4	50.3	2.0*	1.3	-321.4*	-99.4
Total	5849.6	74.2	5841.3	73.0	-8.3	-0.1

Notes: Control weeks are selected through the-neighbor matching using four control observations for each treatment week. Percentiles are relative to the full distribution of control weeks.

- ▶ This would be a good time to work through the vignette for cobalt
<https://cran.r-project.org/web/packages/cobalt/vignettes/cobalt.html>
- ▶ Compare the ATE for the Lalonde data with the IPW, Nearest Neighbor, and Propensity Score estimates.
- ▶ Then start the homework

Difference in Differences

Motivation: Recap Matching

Matching estimators had some advantages:

- ▶ Limited assumptions on **functional forms**
- ▶ We could do nearest neighbor matching and use kernels to compute treatment effects

Matching estimators had some drawbacks:

- ▶ Treated patients were “matched” to control patients based only on **observable characteristics**
 - Ignored **selection on unobservables**.
- ▶ Relied on **cross sectional** variation to construct a control group.

IV estimators resolve some of those issues but

- ▶ Good IV are in short supply!

Often (in this course at least) we have access to **panel data**.

- ▶ What if we could use panel data to control for **unobserved heterogeneity** within a treated individual/group?

Difference in Difference estimators are like the opposite of matching

- ▶ **Strong** assumptions on **functional form**
- ▶ but... allow for **unobservable heterogeneity** in outcomes.

Intro to Difference in Difference

- Consider using OLS to estimate

$$Y_i = \beta_0 + \beta_1 \cdot D_i + \varepsilon_i$$

when $D_i \in \{0, 1\}$ is assigned randomly.

- From the OLS formula and a little algebra, we can show that

$$\hat{b}_1 = \bar{Y}_1 - \bar{Y}_0$$

where $\bar{Y}_1$ is the sample mean of Y_i conditional on $D_i = 1$, and similarly $\bar{Y}_0$ is the sample mean conditional on $D_i = 0$.

- Note: this loops us back to the treatment effects setup, where we were comparing conditional means.

- **Differences-in-differences** is a **quasi-experimental** design that attempts to deal with selection by focusing how treatment and control groups change between two periods
- Consider the model

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{X}_i + \beta_{post} Post_t + \varepsilon_{it},$$

and furthermore suppose that

- Some variables in $\mathbf{X}$ may be unobserved
- D_{it} is not random and may be correlated with $\mathbf{X}_i$
- $Post_t$ is a dummy variable for the post-treatment period $t = 2$
- Each individual i is observed (at least) twice
- For some individuals, D_{it} changes over time (**natural experiment**)
- $\mathbf{X}_i$ does not change over time

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{x}_i + \beta_{post} Post_t + \varepsilon_{it},$$

- Let ΔY_i represent the change in Y_{it} between two time periods:

$$\Delta Y_i = Y_{i2} - Y_{i1},$$

and define ΔD_i and $\Delta \varepsilon_i$ similarly.

- A differenced regression equation:

$$\Delta Y_i = \beta_{post} + \beta_1 \Delta D_i + \Delta \varepsilon_i$$

$$\Delta Y_i = \beta_{post} + \beta_1 \cdot \Delta D_i + \Delta \varepsilon_i$$

- Applying OLS to the differenced regression equation will provide unbiased estimates as long as

$$\mathbb{E} [\Delta \varepsilon_i \mid \Delta D_i] = 0.$$

Exogeneity assumptions of this form are known as the **parallel trends assumption**.

$$\Delta Y_i = \beta_{post} + \beta_1 \cdot \Delta D_i + \Delta \varepsilon_i$$

- ▶ From before, recall that the OLS estimator for a simple experiment amounted to the difference in conditional means of the outcome variable.
- ▶ The same is true here, but now we start with an outcome variable that is differenced over time. Suppose that for some observations, $\Delta D_i = 1$ (treatment group), and for others $D_{i1} = D_{i2} = 0$ (untreated group).

$$\hat{b}_{1,DID} = \overline{\Delta Y}_T - \overline{\Delta Y}_C$$

where $\overline{\Delta Y}_T$ is the conditional mean of ΔY_i for the treatment group, and $\overline{\Delta Y}_C$ is the conditional mean of ΔY_i for the control group.

- ▶ The DID strategy is robust to a form of selection bias (when, the selection is related to persistent characteristics). Simple differences across treatment and control groups would not be.
- ▶ DID estimates are also robust to aggregate shocks or time effects. Simple differences over time for the treatment group would not be.
- ▶ DID can be used to study many “natural experiments,” where something changes for one group but not for an otherwise similar group.

A Famous Example: Card and Krueger (AER 1994)

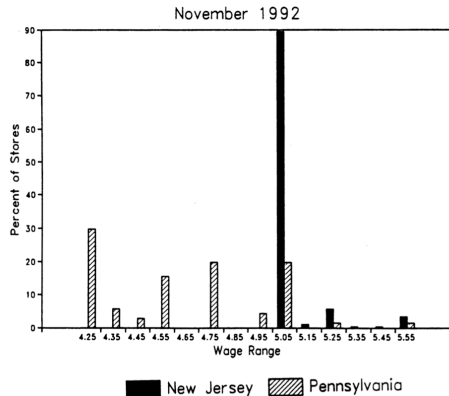
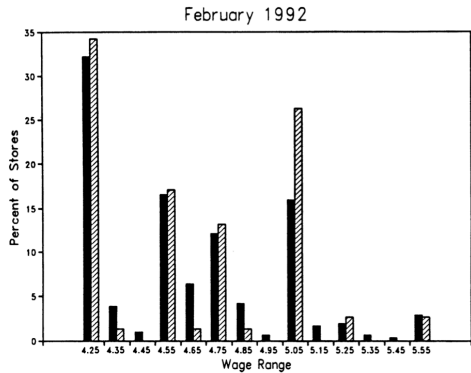
A Famous Example: Card and Krueger (AER 1994)

- ▶ On April 1, 1992 NJ raises its minimum wage from \$4.25 $\rightarrow$ \$5.05 per hour.
- ▶ Question: Econ 101 predicts this will **reduce demand for low wage workers**
 - Focus on fast food restaurants (since they pay min wage)
 - Focus on starting wage (avoid tenure, high turnover)
- ▶ Survey 410 restaurants in NJ (treated group) and eastern PA (control group).
- ▶ Idea: Compare **change** in wages in NJ to PA : $\Delta_{DD} = \Delta_{NJ} - \Delta_{PA}$
 - Wave 1: February 15-March 4, 1992
 - Wave 2: November 5 - December 31, 1992

Balance Table: Covariates

Variable	Stores in:		<i>t</i> ^a
	NJ	PA	
1. <i>Distribution of Store Types (percentages):</i>			
a. Burger King	41.1	44.3	−0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	−1.1
e. Company-owned	34.1	35.4	−0.2
2. <i>Means in Wave 1:</i>			
a. FTE employment	20.4 (0.51)	23.3 (1.35)	−2.0
b. Percentage full-time employees	32.8 (1.3)	35.0 (2.7)	−0.7
c. Starting wage	4.61 (0.02)	4.63 (0.04)	−0.4
d. Wage = \$4.25 (percentage)	30.5 (2.5)	32.9 (5.3)	−0.4
e. Price of full meal	3.35 (0.04)	3.04 (0.07)	4.0
f. Hours open (weekday)	14.4 (0.2)	14.5 (0.3)	−0.3
g. Recruiting bonus	23.6 (2.3)	29.1 (5.1)	−1.0
3. <i>Means in Wave 2:</i>			
a. FTE employment	21.0 (0.52)	21.2 (0.94)	−0.2
b. Percentage full-time employees	35.9 (1.4)	30.4 (2.8)	1.8
c. Starting wage	5.08 (0.01)	4.62 (0.04)	10.8
d. Wage = \$4.25 (percentage)	0.0	25.3 (4.9)	—
e. Wage = \$5.05 (percentage)	85.2 (2.0)	1.3 (1.3)	36.1
f. Price of full meal	3.41 (0.04)	3.03 (0.07)	5.0
g. Hours open (weekday)	14.4 (0.2)	14.7 (0.3)	−0.8
h. Recruiting bonus	20.3 (2.3)	23.4 (4.9)	−0.6

Distribution of Wages



Differences in Wages : 2 x 2 Table

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE
IN NEW JERSEY MINIMUM WAGE

Variable	Stores by state			Stores in New Jersey ^a			Differences within NJ ^b	
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low– high (vii)	Midrange– high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	–2.69 (1.37)	–2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	–2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	–2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores ^c	–2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	–2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0 ^d	–2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	–2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

^aStores in New Jersey were classified by whether starting wage in wave 1 equals \$4.25 per hour ($N = 101$), is between \$4.26 and \$4.99 per hour ($N = 140$), or is \$5.00 per hour or higher ($N = 73$).

^bDifference in employment between low-wage (\$4.25 per hour) and high-wage ($\geq \$5.00$ per hour) stores; and difference in employment between midrange (\$4.26–\$4.99 per hour) and high-wage stores.

^cSubset of stores with available employment data in wave 1 and wave 2.

^dIn this row only, wave-2 employment at four temporarily closed stores is set to 0. Employment changes are based on the subset of stores with available employment data in wave 1 and wave 2.

Outcome Equation

- ▶ Differences lack any covariates (different fast food chains).
- ▶ Also $\Delta_{PA} < 0$ and $\Delta_{NJ} > 0$ (!)
- ▶ Recall i denotes stores, $t \in 1, 2$. Run the following regression:

$$Y_{it} = \beta X_{it} + \alpha \cdot [i \in \text{NJ}] + \gamma \cdot \text{After}_t + \delta \cdot \text{NJ}_i \times \text{After}_t + u_i$$

$$Y_{it} = \beta X_{it} + \alpha \cdot [\text{wage gap}_i] + \gamma \cdot \text{After}_t + \delta \cdot \text{wage gap}_i \times \text{After}_t + u_i$$

- ▶ α is mean difference between NJ and PA
- ▶ γ is mean difference between period 1 and 2
- ▶ δ is the parameter of interest, the **difference in difference**
- ▶ $\text{wage gap}_i = [\min \text{wage}_{i,2} - w_{i1}]_+ = \max\{0, \min \text{wage}_{i,2} - w_{i1}\}$.
(How much do you need to raise $t = 1$ wages to achieve minimum wage in $t = 2$?)

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap ^a	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership ^b	no	yes	no	yes	yes
4. Controls for region ^c	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls ^d	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are -0.237 and 8.825 , respectively. All models include an unrestricted constant (not reported).

^aProportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

^bThree dummy variables for chain type and whether or not the store is company-owned are included.

^cDummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

^dProbability value of joint F test for exclusion of all control variables.

Parallel trends vs pre-trends

- ▶ When the data is available, researchers typically check **pre-trends** as a way of motivating the parallel trends assumption.
 - That is, look at how the control and treatment groups evolve in several periods preceding the time of treatment to see if they were changing in similar ways over time.
- ▶ This is a worthwhile check, and we should certainly worry about the plausibility of the parallel trends assumption if the pre-trends are not parallel.
- ▶ However, the parallel trends assumption is about what *would have happened* immediately at the time of treatment in the absence of treatment, not about what happened before treatment. Thus, establishing parallel pre-trends **does not** establish the parallel trends assumption.
 - Just as we can't test exogeneity within the OLS framework, we can't test the parallel trends assumption within the DiD framework.

Parallel trends vs pre-trends: example

Consider the NJ minimum wage study. Here's a concern that would not be assuaged by seeing parallel pre-trends:

- ▶ If the minimum wage changed because a new administration was elected in NJ and they implemented a portfolio of new policies, then the minimum wage change might have been just one of several ways in which NJ and PA were diverging at the time.
- ▶ Some of the other policy changes might have led to greater labor force participation and/or incentivized businesses to hire more.
- ▶ Because of these other changes, we would expect NJ's restaurant employment to move differently from PA's in 1992 even if the minimum wage change had not happened.
- ▶ This could all be true even if NJ and PA were evolving similarly before 1992. The point is that the treatment was one thing that diverged in 1992, and if one thing was diverging at that point, perhaps other things were also diverging.

- ▶ Recall that before we argued that experimental effects can be estimated with regressions using additional covariates (controls) even though they don't need to be.
- ▶ Similarly, they can be estimated using differences-in-differences rather than simple differences. They don't need to be, as with randomized treatment, the treatment and control groups should have similar baselines.

- ▶ The study has been controversial, and it has had a big impact on policy discussions.
- ▶ Card and Krueger update their conclusion in a 2000 follow up:
The increase in New Jersey's minimum wage probably had no effect on total employment in New Jersey's fast-food industry, and possibly had a small positive effect.
- ▶ There have been *many* criticisms of the paper, including studies showing that the result is not robust to the sample of stores and data source used, and concerns about the external validity of the focus on only fast food stores.
- ▶ Also see studies based on Seattle's 2015 minimum wage change.

A More General Method

Difference in Difference: General Approach

Potential outcome in period 1 = $Y_{i1}(0)$

Potential outcome in period 2 = $\left\{ \begin{array}{ll} Y_{i2}(1) & \text{if } D_{i2} = 1 \\ Y_{i2}(0) & \text{if } D_{i2} = 0 \end{array} \right\}$

	Treatment	Control
Before	$Y_{i1}(0)$	$Y_{i1}(0)$
After	$Y_{i2}(1)$	$Y_{i2}(0)$

We can write the outcome as:

$$Y_{it} = D_{it} \cdot Y_{it}(1) + (1 - D_{it}) \cdot Y_{it}(0) = D_{it} \cdot (Y_{it}(1) - Y_{it}(0)) + Y_{it}(0)$$

Difference in Difference: General Approach

Consider the first difference $\Delta Y_{it} = Y_{i2} - Y_{i1}$:

$$\Delta Y_{it} = D_{i2} \cdot (Y_{i2}(1) - Y_{i2}(0)) + Y_{i2}(0) - Y_{i1}(0)$$

For treated group (first difference):

$$\mathbb{E}[\Delta Y_{it} \mid D_{i2} = 1] = \overbrace{\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid T_{i2} = 1]}^{ATT} + \overbrace{\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 1]}^{\gamma(1)}$$

For control group (second difference):

$$\mathbb{E}[\Delta Y_{it} \mid D_{i2} = 0] = \overbrace{\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 0]}^{\gamma(0)}$$

The DiD (difference in difference) estimator

$$\Delta_{DD} = \mathbb{E}[\Delta Y_{it} \mid D_{i2} = 1] - \mathbb{E}[\Delta Y_{it} \mid D_{i2} = 0]$$

Difference in Difference: Parallel Trends

- ▶ If $\gamma(1) = \gamma(0)$ then the DiD estimator cancels and we are left with the $\Delta_{DD} = \text{ATT}$.
- ▶ This is the **parallel trends assumption**

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 0] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 1]$$

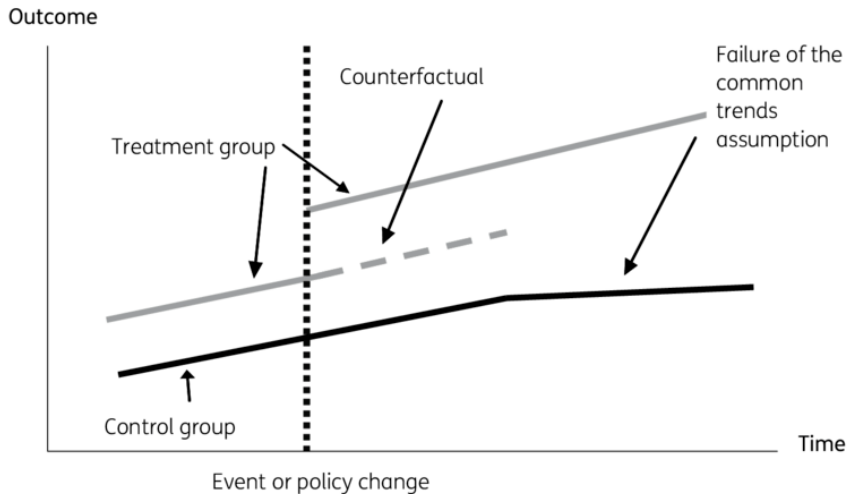
- ▶ Absent the treatment effect, both treatment and control would evolve identically over time.
- ▶ But, treatment and control groups can start from very different places...

$$\mathbb{E}[Y_{it}(0) \mid D_{i2} = 1] \neq \mathbb{E}[Y_{it}(0) \mid D_{i2} = 0], t = 1, 2$$

- ▶ And have selection on treatment effects...

$$\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 1] \neq \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 0]$$

Parallel Trends



Difference in Differences: Limitations

1. Functional form restrictions

- **Parallel trends** assumes that absent treatment that we add $\gamma_2 - \gamma_1$ to each unit
- Because this is **additive** it is not invariant to transformations $f(Y_{it})$ (ie: taking logs)

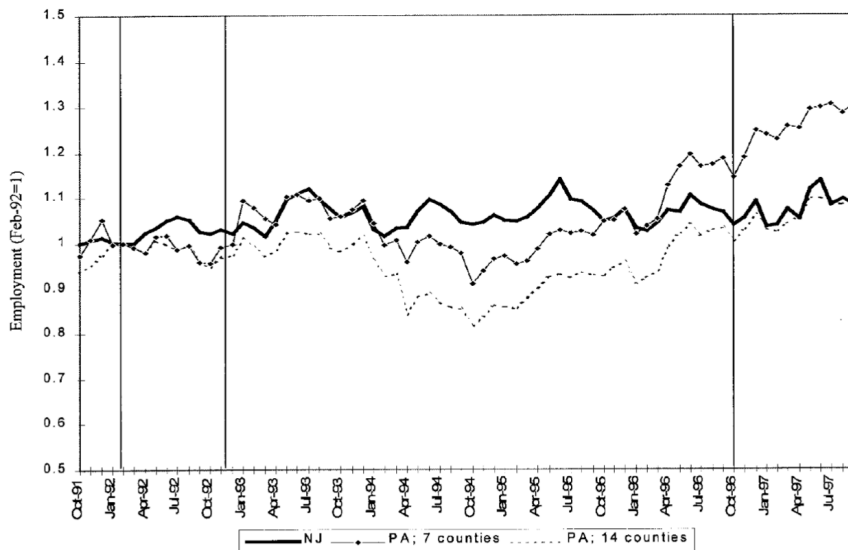
2. Parallel Trend Assumption is **not testable**

- Best we can hope is that it looks similar in the pre-period

3. Compositional Effects: the treatment may affect who is in each group

- Restaurants could close in NJ and open nearby in PA to avoid minimum wage.
- A good job training program may lead to migration, etc.
- One approach: redefine the population so that it doesn't endogenously respond to treatment
 - Recover something, but probably not ATT anymore...

Checking Pre-Trend: Card Krueger (2000)



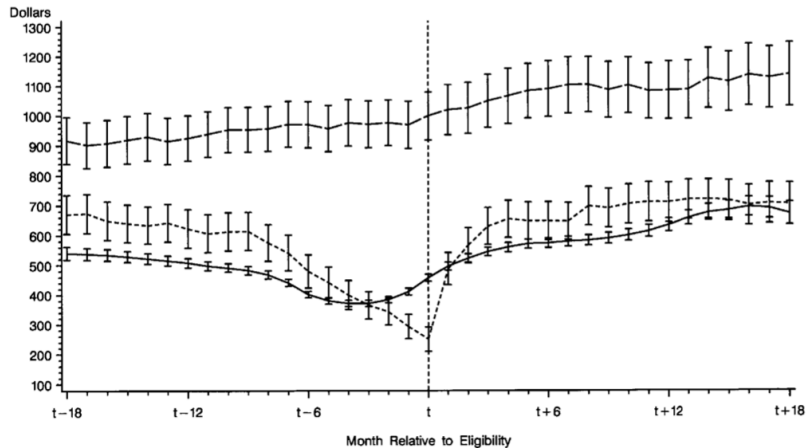
Just like in Card and Kruger, we can write as a regression equation:

$$Y_{it} = \alpha_i + \gamma_t + \beta X_{it} + \delta_i D_{it} + u_{it}$$

- ▶ Suppose we wish to evaluate a training program for those with low earnings. Let the threshold for eligibility be B .
- ▶ We have a panel of individuals and those with low earnings qualify for training, forming the treatment group.
- ▶ Those with higher earnings form the control group.
- ▶ Now the low earning group is low for two reasons
 1. They have low permanent earnings (α_i is low) - this is accounted for by diff in diffs.
 2. They have a negative transitory shock (u_{i1} is low) - this is not accounted for by diff in diffs.

The “Ashenfelter Dip” (Heckman and Smith 2000)

FIGURE 1A
MEAN SELF-REPORTED MONTHLY EARNINGS
SIPP Eligibles and JTPA Controls and ENPs
Male Adults



Difference in Differences

- ▶ #2 above violates the assumption $\mathbb{E}[Y_{i2}(0) - Y_{i1}(0)|T] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0)]$.
- ▶ To see why note that those participating into the program are such that $Y_{i0}(0) < B$. Assume for simplicity that the shocks u are *iid*. Hence $u_{i1} < B - \alpha_i - \gamma_1$. This implies:

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 1] = \gamma_2 = \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} < B - \alpha_i - \gamma_1]$$

For the control group:

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 1] = \gamma_2 = \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} > B - \alpha_i - \gamma_1]$$

So that:

$$\begin{aligned} \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 1] - \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid T = 0] = \\ \mathbb{E}[u_{i1} \mid u_{i1} > B - \alpha_i - \gamma_1] - \mathbb{E}[u_{i1} \mid u_{i1} < B - \alpha_i - \gamma_1] > 0 \end{aligned}$$

- ▶ This is effectively regression to the mean: those unlucky enough to have a bad shock recover and show greater growth relative to those with a good shock. The nature of the bias depends on the stochastic properties of the shocks and how individuals select into training.

- ▶ The assumption on growth of the non-treatment outcome being independent of assignment to treatment may be violated, but it may still be true conditional on X .
- ▶ Consider the assumption

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid X, T] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid X]$$

- ▶ This is just matching assumption on a redefined variable, namely the growth in the outcomes. In its simplest form the approach is implemented by running the regression

$$Y_{it} = \alpha_i + \gamma_t + \delta_i D_{it} + \beta'_t X_i + u_{it}$$

which allows for differential trends in the non-treatment growth depending on X_i . More generally one can implement propensity score matching on the growth of outcome variable when panel data is available.

Variants

The triple difference is also a thing:

- ▶ Suppose that we have: before/after, treated-state/untreated-state, treated-group (men)/untreated-group women.
- ▶ We can compute two D-i-D here: $\Delta_{DDD} = \Delta_{DD,state} - \Delta_{DD,gender}$
- ▶ Literally difference, the difference in differences estimators.
- ▶ As a regression: interpret the triple-interaction term (make sure to control for ALL double interactions).

- Suppose individuals have time-varying observable covariates:

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{x}_{it} + \varepsilon_{it},$$

- We can still estimate β_1 with a differenced linear regression,

$$\Delta Y_i = \beta_1 \cdot \Delta D_i + \beta_2' \Delta \mathbf{x}_i + \Delta \varepsilon_i$$

noting that the exogeneity assumption becomes

$$\mathbb{E} \left[\Delta \varepsilon_i \middle| \begin{pmatrix} \Delta D_i \\ \Delta \mathbf{x}_i \end{pmatrix} \right] = 0$$

- The estimate of β_1 here does not correspond to a simple difference in differences, but people still describe this sort of estimation strategy as DID. Given Frisch-Waugh, it's still DID after controlling for $\mathbf{X}$.

- ▶ A **staggered diff-in-diff** involves different groups being treated at different points in time. Typically, these models are estimated with a linear regression with two-way fixed effects. (More on this when we get to panel data.)
- ▶ See [Callaway and Sant'Anna, de Chaisemartin and D'Haultfoeuille \(2023\)](#) on why this is not a valid strategy in the presence of heterogeneous treatment effects and for alternative estimators. (More on this in Econometrics II)

Synthetic Control

Difference in Difference approaches have some drawbacks:

- ▶ We need to really believe **parallel trends**
 - Is ΔPA really a good counterfactual for ΔNJ ?
 - Obvious question: why not pick ΔDE or ΔNY ?
 - Measured effect shouldn't change if our assumption is valid (but it probably will!)
- ▶ With multiple treated individuals we can use 2WFE estimator

$$y_{it} = \beta X_{it} + \delta_i \cdot T_{it} + \gamma_i + \gamma_t + u_{it}$$

- It assumes **additivity** of FE $\gamma_i + \gamma_t$ are independent!
- States have different baseline levels γ_i but evolution over time is common γ_t .
- Turns out it doesn't really generalize the diff-in-diff with $I > 2$ (Imai and Kim 2020).

Motivation: Recap

Matching Estimators had drawbacks too:

- ▶ We need to believe in as if random-assignment conditional on X
- ▶ CIA: $\{Y_i(1), Y_i(0)\} \perp T_i | X_i$.

But we could use pretty flexible methods in constructing our matched controls:

- ▶ We could try to match on multiple dimensions of X .
- ▶ k -NN, kernels, etc.

What if we could use ideas from **matching** to better satisfy something like our **parallel trends** assumptions?

**Example: Abadie, Diamond,
Hainmueller (2010)**

The Question

In 1988 California passes anti-smoking Prop 99

- ▶ increased excise tax by 25 cents per pack,
- ▶ earmarked the tax revenues to health and anti-smoking education budgets and funded anti-smoking ads
- ▶ led to indoor smoking bans in restaurants and bars city by city

What was the effect on per capita cigarette sales?

- ▶ Already a bunch of pre-existing trends.
- ▶ What is a good control for California?

Use state-level data from 1970-2000.

Use a convex combination of other states to construct a **synthetic counterfactual California**.

- ▶ We observe Y_{it}, X_{it}, D_{it} .
- ▶ Assume only $i = 1$ and $t > T_0$ are **treated**.
- ▶ Construct a **donor pool** of potential controls subscripted by j .
- ▶ Choose some **weights** w_j for each entity (state) in donor pool. How?
 - Same $\mathbf{X}_1 = \sum_j w_j \mathbf{X}_j$ as treated observations (like matching).
 - Same $(Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$ (like parallel trends).
 - Weights sum to one $\sum_j w_j = 1$ and maybe are non-negative (or not!)
- ▶ Idea is to match all of the X 's and all of the Y_{it} 's in the **pre-period**

Table 1. Cigarette sales predictor means

Variables	California		Average of 38 control states
	Real	Synthetic	
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15–24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

NOTE: All variables except lagged cigarette sales are averaged for the 1980–1988 period (beer consumption is averaged 1984–1988). GDP per capita is measured in 1997 dollars, retail prices are measured in cents, beer consumption is measured in gallons, and cigarette sales are measured in packs.

Table 2. State weights in the synthetic California

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Trend Check and Treatment Effects

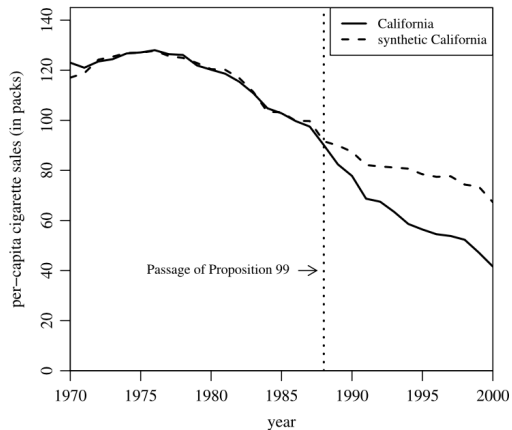
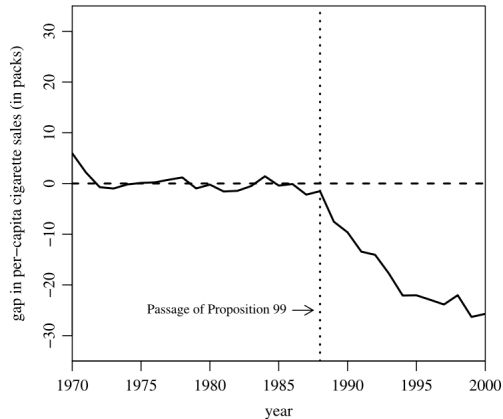


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.



But still some issues

- ▶ How sensitive are weights estimates to different covariates?
 - “state-level measures of unemployment, income inequality, poverty, welfare transfers, crime rates, drug related arrest rates, cigarette taxes, population density, and numerous variables to capture the demographic, racial, and social structure of states”.
- ▶ Can we run a **placebo check**? Do we detect effects where we know there is a null effect?
 - Put California in the donor pool.
 - Pick a state from the donor pool at pretend that receives the treatment after T_0
 - Choose w_j following the synthetic control procedure.
 - Compute the treatment effects in the same way.
 - Repeat for all states in donor pool.
 - Compare **mean-square prediction error** (MSPE) for $(Y_{1,1}, \dots, Y_{1,T_0})$
 - This doubles as **inference**.

Placebo Test

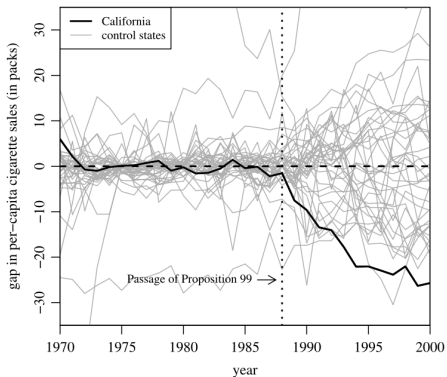


Figure 4. Per-capita cigarette sales gaps in California and placebo gaps in all 38 control states.

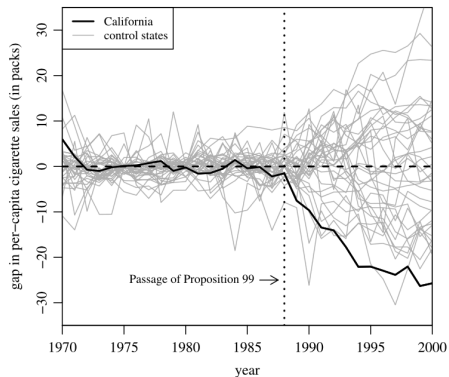


Figure 5. Per-capita cigarette sales gaps in California and placebo gaps in 34 control states (discards states with pre-Proposition 99 MSPE twenty times higher than California's).

More Placebo Tests: How unusual is the CA result?

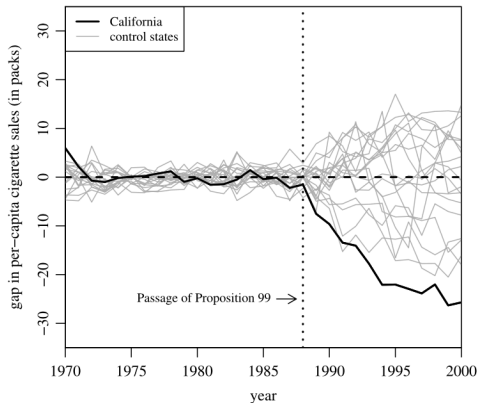


Figure 7. Per-capita cigarette sales gaps in California and placebo gaps in 19 control states (discards states with pre-Proposition 99 MSPE two times higher than California's).

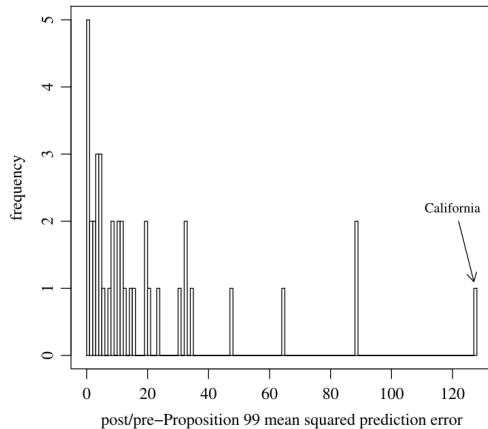


Figure 8. Ratio of post-Proposition 99 MSPE and pre-Proposition 99 MSPE: California and 38 control states.

We're going to break this down into three parts:

1. How do we estimate w_j ?
2. Conditional on $\hat{w}_j$'s, how do we estimate the treatment effect?
3. What kinds of models are compatible?

Start with $D_{it} = 1$ IFF $i = 1$ and $t > T_0$:

$$Y_{it} = Y_{it}(0) + \alpha_{it} \cdot D_{it}$$

Estimate $\alpha_1 = (\alpha_{1,T_0+1}, \alpha_{1,T_0+2}, \dots, \alpha_{1,T})$ (period by period treatment effect).

For $t > T_0$:

$$\alpha_{1t} = Y_{1t}(1) - Y_{1t}(0) = \underbrace{Y_{1t}}_{\text{observed}} - \underbrace{Y_{1t}(0)}_{\text{counterfactual}}$$

Consider the model

$$Y_{it}(0) = \gamma_t + \theta_t \mathbf{Z}_i + \lambda_t \mathbf{F}_i + u_{it}$$

- ▶ δ_t is the time fixed effect
- ▶ Z_i are the usual covariates (fixed in i over time) with potentially time varying coefficients.
- ▶ F_i are i specific **unobserved factors**. Estimating these are complicated.
- ▶ λ_t are called **factor loadings**.

This is a generalization of the usual additive $\gamma_t + \gamma_i$ fixed effects model called **interactive fixed effects** (Bai 2009).

So we define the synthetic observation as:

$$\sum_{j=2}^{J+1} w_j \underbrace{Y_{jt}(0)}_{\text{observed}} = \delta_t + \theta_t \left(\sum_{j=2}^{J+1} w_j \mathbf{z}_j \right) + \lambda_t \left(\sum_{j=2}^{J+1} w_j F_j \right) + \sum_{j=2}^{J+1} w_j u_{jt}$$

And the difference in untreated observations:

$$\begin{aligned} & Y_{1t}(0) - \sum_{j=2}^{J+1} w_j Y_{jt}(0) \\ &= \underbrace{\theta_t \left(\mathbf{z}_1 - \sum_{j=2}^{J+1} w_j \mathbf{z}_j \right)}_{\text{match this directly}} + \underbrace{\lambda_t \left(F_1 - \sum_{j=2}^{J+1} w_j F_j \right)}_{\text{match indirectly: } (Y_{i,1}, \dots, Y_{i,T_0})} + \sum_{j=2}^{J+1} w_j \underbrace{(u_{1t} - u_{jt})}_{\text{Usual } \mathbb{E}[u_{it} | \mathbf{z}_{it}, \mathbf{F}_{it}] = 0} \end{aligned}$$

The goal is to match:

$$\mathbf{z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{z}_j, \quad \mathbf{F}_1 \approx \sum_{j=2}^{J+1} w_j \mathbf{F}_j$$

- The main result in the Abadie et. al (2010) paper proves that the above equation holds approximately if

$$(Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$$

- This frame generalizes the 2WFE model since we can use time dimension to difference them out.
- We cannot difference out the $\lambda_t \mathbf{F}_i$.

The Challenge

Theory depends on matching:

$$\mathbf{z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{z}_j, \quad (Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$$

- ▶ One problem is the **convex hull** problem.
 - There may be no $w_2, \dots, w_J$ that satisfy the equations
- ▶ Instead choose $\mathbf{w}$ in a **minimum distance sense** with some weighting matrix Ω
 - Can choose Ω to minimize variance of the resulting estimator.
- ▶ Hard to fit crazy out of bounds values: can't match CA population as linear combination of other states!

Other things can go wrong

$$\mathbf{x}_i = (\mathbf{z}_i, Y_{i,1}, \dots, Y_{i,T_0})$$
$$\arg \min_{w_2, \dots, w_J} (\mathbf{x}_i - \sum_j w_j \mathbf{x}_j)' \Omega (\mathbf{x}_i - \sum_j w_j \mathbf{x}_j) \quad \text{s.t.} \quad \sum_j w_j = 1$$

- ▶ Can get negative weights: what do those mean?
- ▶ We can rule them out $w_j \geq 0$ constraints but the problem becomes more difficult to solve.
- ▶ This starts to look like a familiar LASSO problem: quadratic objective, L_1 penalty!
- ▶ We shouldn't be surprised by sparse models.

- ▶ The R package `synth` or `tidysynth` and accompanying examples are a good start.
- ▶ The R package `MSCMT` fixes some of the optimization issues in `synth`.
 - When there is no set of feasible weights and the non-negativity constraint binds – the problem is tricky to solve
 - `synth` can produce sub-optimal solutions (and slowly).

We can ask – what are the implied regression weights for linear regression? What is $\mathbb{E}[Y_{1,t}(0) \mid \mathbf{x}]$?

- ▶ Get coefficients using just donor pool: $\widehat{B} = \left(\overline{X}_0 \overline{X}_0'\right)^{-1} \overline{X}_0 Y_0'$
- ▶ Predict $Y_{1,t}(0) = \widehat{B}' \overline{X}_1$.
- ▶ Same as synthetic control but with weights: $W^{reg} = \overline{X}_0' \left(\overline{X}_0 \overline{X}_0'\right)^{-1} \overline{X}_1$ (Projection matrix).

Why? Remember, **everything is a kernel** our prediction of $\mathbb{E}[Y \mid X]$ is always some weighted average of other Y_j 's.

- ▶ What is word of mouth effect of Superbowl ads? (Lovett, Peres, Xu QME 2019)
 - Construct synthetic versions of brands that don't buy Superbowl ads
- ▶ What is effect of soda tax on consumption in Berkeley, CA ? (Bollinger Sexton WP 2019).
 - Construct synthetic drugstores, supermarkets, and convenience stores using two donor polls: inside CA and outside CA.
- ▶ What are economic effects of German reunification?
 - Now we need a counterfactual "West Germany" (!)
(Abadie, Diamond, and Hainmueller 2015)

TABLE 1 Synthetic and Regression Weights for West Germany

Country	Synthetic Control Weight	Regression Weight	Country	Synthetic Control Weight	Regression Weight
Australia	0	0.12	Netherlands	0.09	0.14
Austria	0.42	0.26	New Zealand	0	0.12
Belgium	0	0	Norway	0	0.04
Denmark	0	0.08	Portugal	0	−0.08
France	0	0.04	Spain	0	−0.01
Greece	0	−0.09	Switzerland	0.11	0.05
Italy	0	−0.05	United Kingdom	0	0.06
Japan	0.16	0.19	United States	0.22	0.13

Notes: The synthetic weight is the country weight assigned by the synthetic control method. The regression weight is the weight assigned by linear regression. See text for details.

**TABLE 2 Economic Growth Predictor Means
before German Reunification**

	West Germany	Synthetic West Germany	OECD Sample
GDP per capita	15808.9	15802.2	8021.1
Trade openness	56.8	56.9	31.9
Inflation rate	2.6	3.5	7.4
Industry share	34.5	34.4	34.2
Schooling	55.5	55.2	44.1
Investment rate	27.0	27.0	25.9

Notes: GDP per capita, inflation rate, trade openness, and industry share are averaged for the 1981–90 period. Investment rate and schooling are averaged for the 1980–85 period. The last column reports a population-weighted average for the 16 OECD countries in the donor pool.

FIGURE 1 Trends in per Capita GDP: West Germany versus Rest of the OECD Sample

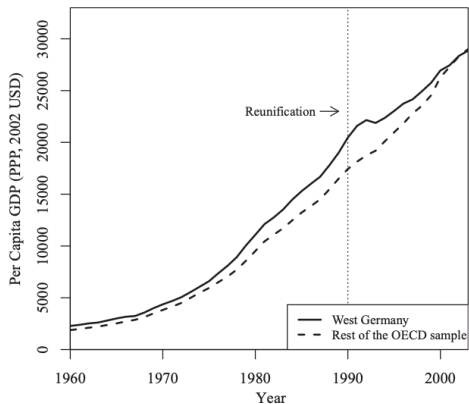


FIGURE 2 Trends in per Capita GDP: West Germany versus Synthetic West Germany

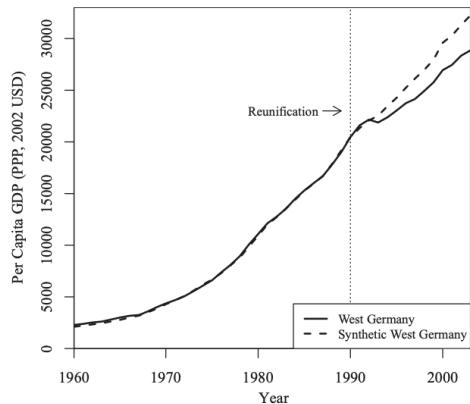


FIGURE 3 Per Capita GDP Gap between West Germany and Synthetic West Germany

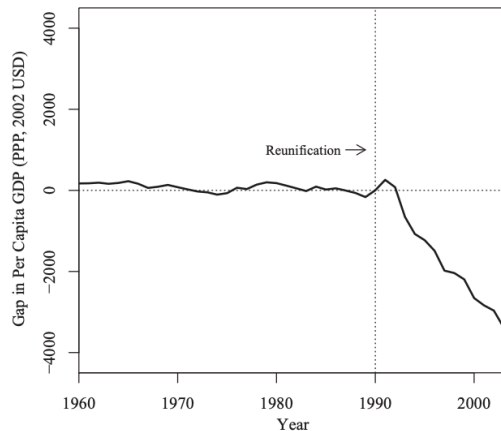


FIGURE 4 Placebo Reunification 1975—Trends in per Capita GDP: West Germany versus Synthetic West Germany

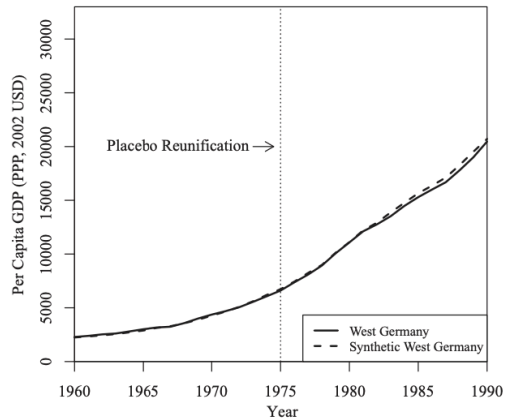
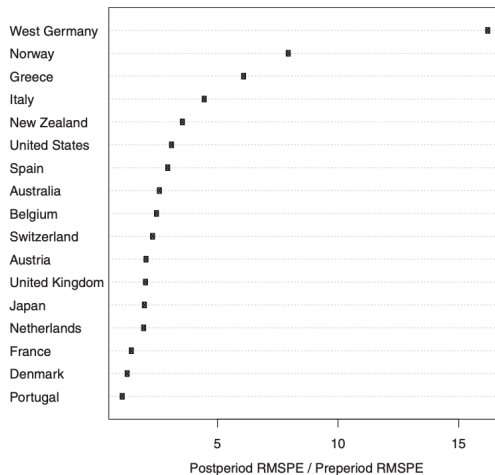


FIGURE 5 Ratio of Postreunification RMSPE to Prereunification RMSPE: West Germany and Control Countries



Thanks
